Tahereh Toosi

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Appointments

Center for Theoretical Neuroscience, Columbia University — New York, NY
Associate Research Scientist — 2022 – Present
Mentors: Prof. Kenneth D. Miller and Prof. Larry F. Abbott
Supported by NIH pathway to independence award (parent K99/R00) from NEI

Zuckerman Mind Brain Behavior Institute, Columbia University — New York, NY
Postdoctoral Research Scientist — 2018 – 2022
Advisor: Dr. Elias B. Issa

Education

Institute for Research in Fundamental Sciences (IPM) — Tehran, Iran
PhD in Cognitive Sciences — 2017
Advisor: Prof. Hossein Esteky
Thesis: Effect of Temporal Context on Visual Sensory Processing

Australian National University — Canberra, Australia
Visiting graduate student — 2015 – 2016
Advisor: Prof. Ehsan Arabzadeh

Massachusetts Institute of Technology — Cambridge, MA
Visiting graduate student — 2014 – 2015
Mentor: Prof. James J. DiCarlo
Advisors: Dr. Arash Afraz and Dr. Daniel Yamins

Iran University of Science and Technology — Tehran, Iran
MSc in Biomedical Engineering — 2009
Advisor: Prof. Hamid Behnam
Thesis: Optimization of Ultrasound Array Beamforming for Medical Applications

Ferdowsi University of Mashhad — Mashhad, Iran
BSc in Electrical Engineering — 2006
Advisor: Prof. Reza Lotfi
Thesis: A CAD Tool for Designing High-Performance Op-Amps

Publications

Google Scholar link here.

Toosi, T., & Issa, E. B. (2023). Brain-like flexible visual inference by harnessing feedback-feedforward alignment. *Advances in Neural Information Processing Systems (NeurIPS).*

Toosi, T., & Issa, E. B. (2023). Brain-like temporal straightening of natural movies in robust feedforward neural networks. In *International Conference on Learning Representations (ICLR).*

Toosi, T. (2023). Representational constraints underlying similarity between task-optimized neural systems. In *UniReps Workshop, Neural Information Processing Systems (NeurIPS).*

Toosi, T., Kriegeskorte, N., & Issa, E. B. (2023). Object-enhanced and object-centered representations across primate ventral visual cortex. In *Cognitive Computational Neuroscience (CCN).*

Kell, A. J. E., Bokor, S., Jeon, Y., **Toosi, T.**, & Issa, E. B. (2023). Marmoset core visual object recognition behavior is comparable to that of macaques and humans. *iScience*.

Toosi, T. (2022). Perceptually-aligned gradients by sampling the implicit prior. In *DeepMath*.

Toosi, T., & Issa, E. B. (2020). Symbiotic learning of dual discrimination and reconstruction networks. In *Beyond Backpropagation workshop, Neural Information Processing Systems (NeurIPS)*.

Toosi, T., Tousi, E. K., & Esteky, H. (2016). Learning temporal context enhances the prestimulus alpha oscillations in the parietal cortex and improves the visual discrimination performance. *Journal of Neurophysiology*.

Toosi, T. K., & Behnam, H. (2009). Combined pulse compression and adaptive beamforming in coded excitation ultrasound medical imaging. In *Intl. Conference of Signal Processing Systems*.

Sarbishaei, H., **Toosi, T. K.**, Zhian-T., E., & Lotfi, R. (2006). Power optimization in low-voltage high-speed high-resolution pipelined ADCs. In *Midwest Symposium on Circuits and Systems (MWSCAS)*.

Toosi, T. K., Zhian-T., E., Sarbishaei, H., & Lotfi, R. (2006). ISECAD: An iterative simulation-equation-based opamp-design CAD tool. In *Intl. Symp. Circuits and Systems (ISCAS)*.

Sarbishaei, H., **Toosi, T. K.**, Zhian-T, E., & Lotfi, R. (2005). A high gain high speed low power class AB operational amplifier. In *Midwest Symposium on Circuits And Systems (MWSCAS)*.

Talks

Emergence of illusory contours in robust deep neural networks by accumulation of implicit priors
Object Recognition: Models, Vision Science Society Meeting, St. Pete Beach Florida, 2024

Can images predict neural patterns better than Deep Nets?
ICBINB Workshop, Cosyne Meeting, Lisbon, Portugal, 2024

Harnessing feedback pathways: Integrating perceptual priors in sensory processing
SYNAPSES Seminar Series, Yale University, 2024

Cortical computations underlying the integration of perceptual priors and sensory processing
Brain Science External Postdoc Seminar Series, Brown University, 2024

Uncovering the evolution of neural representations in the ventral visual stream
Neuroscience and Artificial Intelligence Laboratory (NeuroAILab), Stanford University, 2023

Interpretable intermediate representations in primate ventral visual cortex
Visual Inference Lab, Columbia University, 2023

Representational straightening of natural movies in robust feedforward neural networks
Visual Object and Scene Recognition Nanosymposium, Society for Neuroscience Meeting, San Diego, 2022

Symbiotic learning of feedforward and feedback networks
from Neuroscience to Artificially Intelligent Systems (NAISys), Cold Spring Harbor Labratory, 2020

Teaching Experience

Invited Lecturer **Advanced Topics in Theoretical Neuroscience**, 2024
Deep neural networks as models in neuroscience

Mentor **Neuromatch Academy**, 2023
Deep learning: computer vision

Instructor **Methods in Neuroscience Workshop at IPM**, 2013
Statistical approaches to neural data modeling, Methods of firing rate estimation, Information theory approaches

Instructor **IBRO School of Neuroscience**, 2011
EEG/ERP experiment design and recording

Awards and Honors

Travel Grant for Highly Scored Abstract **CoSyNe Meeting**, 2024

Pathway to Independence Award - parent K99/R00 **NIH/NEI**, 2023
Computational and Theoretical Understanding of Regulatory Mechanisms Shaping Natural Vision

Visiting Graduate Student Scholarship **IPM and MIT**, 2014-2015
McGovern Institute at MIT

Scholarship **Cold Spring Harbor Laboratory Asia**, 2012
Summer School on Computational and Cognitive Neuroscience

Scholarship **FENS/IBRO**, 2011
Advanced Course in Computational Neuroscience, Summer School

Distinguished Researcher Award **Ferdowsi University**, 2005

Honored at the National Mathematics Olympiad 1997 and 1998

Ranked 1st at the Province Computer Olympiad 1997